RAJALAKSHMI ENGINEERING COLLEGE

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P2

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|---|---|
| **Started on** | Wednesday, 26 August 2026, 8:50 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 26 August 2026, 8:52 PM |
| **Time taken** | 2 mins 7 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>

void quickSort(int arr[], int low, int high)
{
    if (low >= high)
        return;
    int i = low;
    int j = high;
    int pivot = arr[(low + high) / 2];
    while (i <= j)
    {
        while (arr[i] < pivot)
            i++;
        while (arr[j] > pivot)
            j--;
        if (i <= j)
        {
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
            i++;
            j--;
        }
    }
    quickSort(arr, low, j);
    quickSort(arr, i, high);
}

int main()
{
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    quickSort(arr, 0, n - 1);
    for (int i = 0; i < n; i++)
    {
        if (i > 0)
            printf(" ");
        printf("%d", arr[i]);
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 | |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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